Important Tables

The tables from the textbook, plus comparison tables built from its text. Each table says where it comes from.



Chapter 1 · Respiration

1.1 Parts of the respiratory system and what they do

Part	Also called	What it does
Trachea	windpipe	Carries air from the nose and mouth towards the lungs. Rings of cartilage keep it open.
Larynx	voicebox	Wide part at the top of the trachea. Contains the vocal cords, which vibrate to make sound.
Bronchus (pl. bronchi)		Two branches of the trachea, one to each lung. Also supported by cartilage.
Bronchiole		Smaller tubes from each bronchus; carry air deeper into the lungs.
Air sac	alveolus (pl. alveoli)	Where oxygen goes into the blood and carbon dioxide comes out.
Intercostal muscles	muscles between the ribs	Contract to pull the ribs up and out when breathing in.
Diaphragm		Muscle under the lungs. Contracts and moves down when breathing in; relaxes back to a dome when breathing out.

Built from the text of Topics 1.1 and 1.3.

1.2 Inspired air and expired air

	Inspired air (breathed in)	Expired air (breathed out)
Oxygen	a lot	less (some went into the blood)
Carbon dioxide	only a small amount	more (made by respiring cells)
Limewater test	stays clear longer	turns limewater cloudy first

Built from the text of Topic 1.2. The textbook gives no percentages, so none are shown.

1.3 Breathing in and breathing out

	Breathing in	Breathing out
Diaphragm muscles	contract	relax
Intercostal muscles	contract	relax
Diaphragm	pulled downwards	back to its normal domed shape
Ribs	pulled upwards and outwards	drop down to their natural position
Volume of chest	increases	decreases
Pressure in chest and lungs	decreases	increases
Air	moves into the lungs	is squeezed out of the lungs

First two rows: the textbook table in Topic 1.3 (answers filled in). Other rows: from the text of Topic 1.3.

1.3 Syringe model of breathing

Part of the model	Part of the body
Balloon	lungs
Small hole in the rubber bung	trachea
Plunger	diaphragm
Syringe barrel	rib cage (does not move, unlike real ribs)

Answers to the Topic 1.3 model questions.

1.4 Aerobic respiration

Reactants	Products	Where it happens	What is released
glucose, oxygen	carbon dioxide, water	mitochondria, in every living cell	energy (a little at a time); some becomes heat

`glucose + oxygen → carbon dioxide + water`

From the text of Topic 1.4.

1.4 Breathing and respiration are different

	Breathing	Respiration
What it is	movements of the diaphragm and intercostal muscles	a series of chemical reactions
Where	chest and lungs	inside every living cell (in mitochondria)
What it does	moves air into and out of the lungs	releases useful energy from glucose

From Topics 1.3 and 1.4 (Activity 1.4.2).

1.4 Respiration in peas: the variables

Variable	In this experiment
Changed	dead peas or live, germinating peas
Measured	temperature inside each flask (at regular intervals, e.g. every hour)
Kept the same	everything else: same number of peas, flasks, cotton wool, insulation, thermometers
Result	the flask of live peas gets warmer, because respiring cells release heat

Answers to the Topic 1.4 investigation questions.

1.5 Components of blood

Component	Appearance	Function
Red blood cell	red; small; no nucleus, no mitochondria; full of haemoglobin	carries oxygen (as oxyhaemoglobin) to respiring cells
White blood cell	always has a nucleus; some kinds are larger than red cells	defends the body against pathogens (captures them, or makes antibodies)
Plasma	liquid; very pale yellow; mostly water	transports blood cells and dissolved substances such as glucose and carbon dioxide

The textbook table in Topic 1.5 (question 4), filled in from the text.

Chapter 2 · Properties of materials

2.1 Dissolving or melting?

Examples of dissolving	Examples of melting
Sugar (solute) in black tea (solvent)	Butter in a frying pan
Instant coffee (solute) in hot water (solvent)	Ice cream on a warm day
Nail polish (solute) in nail polish remover (solvent)	Candle wax as the candle burns

Dissolving needs two substances, a solute and a solvent.

Textbook table, Topic 2.1.

2.2 Solutes and solvents in these chapters

Solute	Solvent	Notes
sugar, salt (sodium chloride)	water	colourless, transparent solutions
copper sulfate	water	blue, but transparent
oil paint	methanol (methylated spirits)	not soluble in water
nail polish	propanone (acetone)	not soluble in water
permanent marker ink	alcohol	not soluble in water
green pigment from spinach	ethanol	used for chromatography

Built from the text of Topics 2.1, 2.2 and 2.4.

2.2 Making different concentrations of food dye

Solution	Concentrated dye (cm^3)	Water (cm^3)	Total volume (cm^3)
A	10	0	10
B	8	2	10
C	6	4	10
D	4	6	10
E	2	8	10

A is the most concentrated (most dye particles); E is the most dilute (palest).

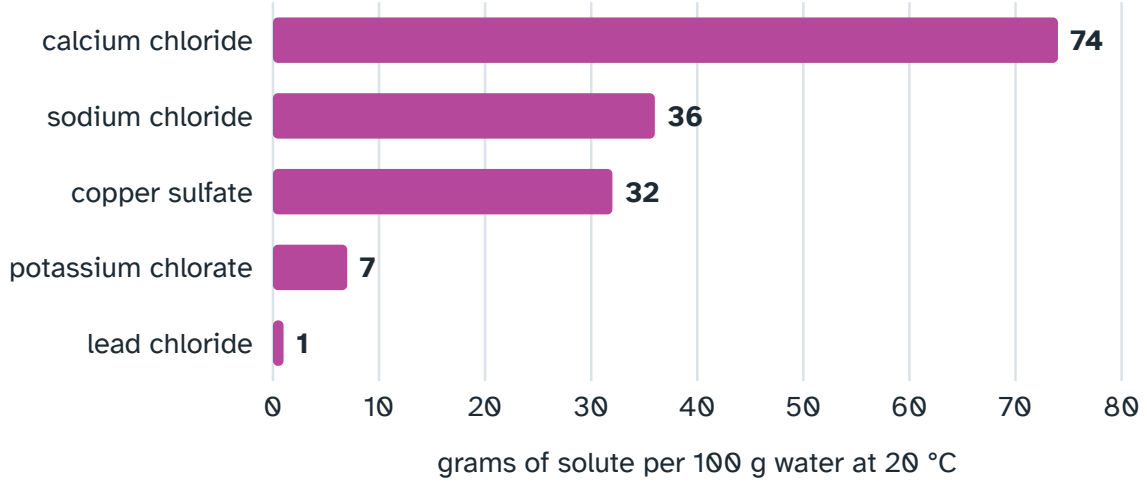
Textbook table, Topic 2.2.

2.2 Solubility of salts in water at 20 °C

Solute	Solubility (g per 100 g water at 20 °C)
calcium chloride	74
sodium chloride	36
copper sulfate	32
potassium chlorate	7
lead chloride	1

Most soluble: calcium chloride. Least soluble: lead chloride. Example: potassium chlorate in 200 g water = 7 × 2 = 14 g; sodium chloride in 50 g water = 36 ÷ 2 = 18 g.

Textbook table, Topic 2.2 (shown here from most to least soluble).



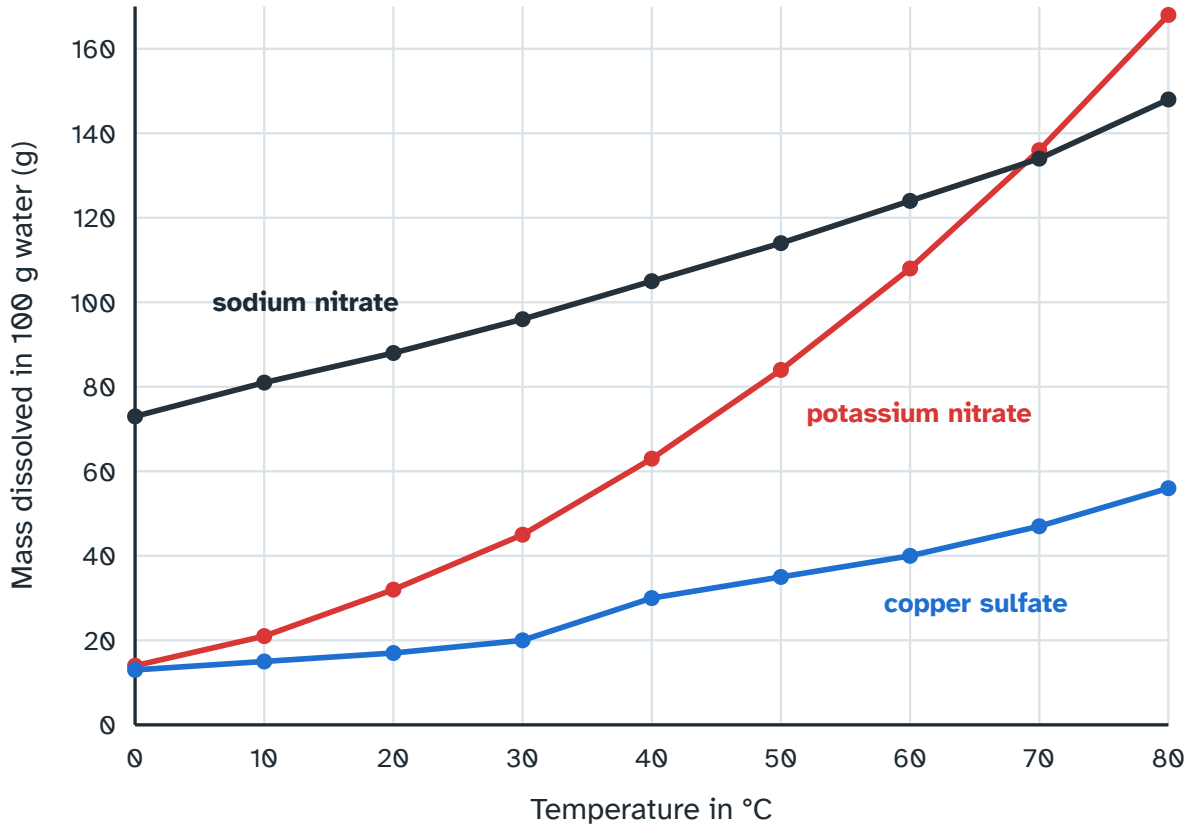
2.2 Solubility of three salts at different temperatures

Temperature (°C)	Potassium nitrate (g/100 g water)	Sodium nitrate (g/100 g water)	Copper sulfate (g/100 g water)
0	14	73	13
10	21	81	15
20	32	88	17
30	45	96	20
40	63	105	30
50	84	114	35
60	108	124	40
70	136	134	47
80	168	148	56

All three get more soluble as temperature rises. Most soluble at 10 °C: sodium nitrate. Most soluble at 80 °C: potassium nitrate.

Note: this table gives copper sulfate as 17 g at 20 °C, but the 20 °C table above gives 32 g. Both are printed in the textbook; ask the teacher which to learn.

Textbook table, Topic 2.2.



2.2 Sugar: solubility and temperature

Mass of water	Sugar dissolved at 20 °C	Sugar dissolved at 80 °C
100 g	204 g	362 g
200 g	408 g	724 g*
50 g	102 g	181 g*
250 g	510 g	905 g (395 g more than at 20 °C)

From the worked example and questions 6–7 in Topic 2.2. Method: find the amount for 1 g of water (204 ÷ 100 = 2.04 g), then multiply.
*Worked out the same way from 362 g per 100 g; not printed in the book.

2.2 Check your progress 2.2: four compounds

Name	Colour	Solubility in water
sodium chloride	white	soluble
zinc carbonate	white	insoluble
iron sulfate	green	soluble
copper carbonate	green	insoluble

After filtering, zinc carbonate is the white solid left in the filter paper; its filtrate is colourless.

Textbook table, Check your progress 2.2.

2.4 Reading a chromatogram

What you see	What it means
One spot	a pure substance
Two or more spots	a mixture of substances
Spot carried far up the paper	very soluble in the solvent
Spot stays near the pencil line	not very soluble
Spot matches a known dye at the same height	probably the same substance
Spot matches no permitted dye	not a permitted colouring

Built from the text and questions of Topic 2.4.

Also tested in Worksheet 2

WS2 Types of solution

Solution	Solute	Solvent	Type
Sweet tea	sugar	water	solid in a liquid
Sea water	salt	water	solid in a liquid
Soda water / fizzy drink	carbon dioxide	water	gas in a liquid
Brass (an alloy)	zinc	copper	solid in a solid

Worksheet 2, Q14 and Q16 (answer key).

WS2 Dilute, concentrated, saturated, unsaturated

Term	Meaning
Dilute	a small amount of solute in a lot of solvent
Concentrated	a large amount of solute in the solvent
Saturated	no more solute can dissolve at that temperature
Unsaturated	more solute can still dissolve
Aqueous	water is the solvent

Worksheet 2, Q13 and Q19 (answer key).

WS2 Separating a mixture of sand, salt and water

Step	Method	What you get
1	Filtration	sand stays in the filter paper (residue); salt solution passes through (filtrate)
2	Wash and dry the residue	dry sand
3	Evaporation: heat the filtrate in an evaporating dish	salt crystals

Worksheet 2, Q7 and Q11 (answer key).

WS2 Time for 5 g of sugar to dissolve in 100 cm³ of water

Temperature (°C)	Time to dissolve (s)
20	120
40	75
60	45
80	25

Independent variable: temperature. Dependent: time. Keep the same: mass of sugar, volume of water, stirring, crystal size. Hotter → dissolves faster. At 70 °C expect about 30–40 s.

Worksheet 2, Q10.

WS2 Solubility curves (g per 100 g of water)

Temperature (°C)	Potassium nitrate	Sodium chloride	Potassium chloride
0	13	36	28
10	21	36	31
20	32	36	34
30	46	36	37
40	64	37	40
50	85	37	43
60	110	37	46
70	138	38	48
80	169	38	51
90	202	39	54
100	246	40	56

Read from Worksheet 2, Figure 2; the worksheet accepts answers within ±3 g. Crystals formed when 100 g of water saturated with potassium nitrate cools from 60 °C to 20 °C: 110 − 32 ≈ 78 g.

Values read from the Worksheet 2 graph, not printed as a table in the worksheet.

WS2 Solubility of solid Z

Temperature (°C)	0	20	40	60	80
Solubility (g per 100 g water)	10	16	25	39	60

From the graph: about 31–32 g at 50 °C. At 40 °C, Z (25 g) is less soluble than potassium nitrate (64 g).

Worksheet 2, Q29.

WS2 Worksheet 2 chromatogram: Rf values

Sample	Spots (cm from baseline)	Rf value	Pure?
Dye A	3.0	0.30	yes
Dye B	5.0	0.50	yes
Dye C	7.5	0.75	yes
Ink X	3.0 and 7.5		no: dyes A and C
Ink Y	5.0 and 7.5		no: dyes B and C

Rf = distance moved by the spot ÷ distance moved by the solvent front (10.0 cm). Rf is always less than 1.

Worksheet 2, Figure 4 and Q32–Q35. Rf values worked out from the figure; the worksheet lists Rf as a key word but does not ask for the values.